

GE-Problem Set 1, ECN 714

Exercise 1 A certain gourmand claims that it is impossible to consume an ounce of chili (a local delicacy) without the accompaniment of between $\frac{1}{2}$ and 2 ounces of beer. The exact quantity consumed is a matter of individual preference.

Similarly, beer can only be consumed with chili. Assume that chili can be consumed in any nonnegative amount.

- Draw the individual's consumption set.
- Consider a two person exchange economy in which each of the agents has consumption sets as in part (a). What does an Edgeworth box look like for this economy?
- Show by example that there are economies of the type described in (b) in which utility is *strictly* increasing and there are equilibria in which one of the goods has price zero. What must happen for this to be true?
- Can the result in (c) occur when the consumption set is the positive orthant?

Exercise 2 Consider an exchange economy with two consumers and two goods. Goods can be consumed only in integer, non-negative quantities. Assume that both consumers have continuous, strictly monotone and strictly concave utility functions.

- Is every Walrasian equilibrium Pareto efficient? Prove it or found a counter-example.
- Suppose now that one of the two commodities is perfectly divisible. Is every Walrasian equilibrium Pareto efficient? Prove it or found a counter-example.

Exercise 3 Consider a private ownership economy with one consumer, one firm and two commodities, food F (measured in pounds per day) and labor L (measured in hours per day). The consumer has no initial endowment of either good and can work up to 24 hours a day. For cases A and B specified below, please answer questions (a), (b), and (c). You may ignore corner solutions in (b) and (c).

Case A: The utility function and the firm's technology are

$$U(F, L) = \log F + \log(24 + L)$$

$$F = (-L)^{0.5}.$$

Case B: The utility function is $U(F, L) = \log F + L$. The firm's technology is constant returns to scale and satisfies two conditions. The first condition is that a person-hour of labor (in absolute terms) yields a pound of food. The second condition is that for every hour that the agent works, the agent needs 0.5 pound of food to survive.

- a. Draw a graph depicting this private ownership economy, including the consumer's consumption set, preference relation and the firm's technology.
- b. Calculate the Pareto efficient allocations. Show them on the graph in (a).
- c. Calculate the Walrasian equilibria. Show them on the graph in (a).